STPM Physics · Semester 1 · Chapter 10

Gases

10.1 Ideal Gas Equation

Definitions & laws

- **Ideal gas** is a gas that obeys the equation of state pV = nRT at all pressures and temperatures.
- **Boyle's law**: for a fixed mass of gas at constant temperature, the pressure is inversely proportional to the volume. **Charles's law**: at constant pressure, the volume is directly proportional to the thermodynamic temperature. **Pressure law**: at constant volume, the pressure is directly proportional to the thermodynamic temperature.
- **Mole** is the amount of substance that contains as many elementary entities as there are atoms in 0.012 kg of carbon-12, i.e. the Avogadro constant N_A of entities.
- A real gas behaves like an ideal gas at low pressure and high temperature (molecules far apart, intermolecular forces and molecular volume negligible).
- **Dalton's law**: the total pressure of a mixture of gases equals the sum of the partial pressures of its components.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Ideal gas equation (moles)	$pV = nRT$	p = pressure (Pa); V = volume (m^3); n = amount of substance (mol); R = 8.31 J mol^{-1} K^{-1}; T = thermodynamic temperature (K)
Ideal gas equation (molecules)	$pV = NkT$	N = number of molecules; $k = \frac{R}{N_A} = 1.38\times10^{-23}$ J K^{-1} (Boltzmann constant); T = thermodynamic temperature (K)
Amount of substance	$n = \frac{N}{N_A} = \frac{m}{M}$	$N_A = 6.02\times10^{23}$ mol^{-1}; m = mass (kg); M = molar mass (kg mol^{-1})
Gas law for fixed mass	$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$	for a fixed mass of gas; T in kelvin
Temperature conversion	$T\ (K) = \theta\ (°C) + 273$	T = thermodynamic temperature (K); θ = Celsius temperature (°C); 273.15 may be used instead of 273
Density of an ideal gas	$\rho = \frac{pM}{RT}$	from $pV = \frac{m}{M}RT$; ρ = density (kg m^{-3}); M = molar mass (kg mol^{-1})

How to answer (MPM style)

- Write the equation first (pV = nRT), then convert: temperature to K, volume to m^3 (1 cm^3 = 10^{-6} m^3, 1 litre = 10^{-3} m^3), pressure to Pa.
- Make the quantity the subject, substitute with units, and give the answer with unit to 3 s.f.
- For a change of state of a fixed mass, use $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$ and state which quantity is constant when a law is quoted.
- Explanation of deviation: state that real gases deviate at high pressure and low temperature because the molecular volume and intermolecular forces are no longer negligible.

Section A traps

Trap: Substituting the temperature in °C instead of K; the result can be wrong by a large factor and sometimes gives negative values.

Trap: Using the molar mass in g mol^{-1} with R = 8.31 J mol^{-1} K^{-1}, which requires kg mol^{-1}; or using the volume in cm^3 or litres without conversion to m^3.

Trap: Mixing up R and k: R is used with the number of moles n, k with the number of molecules N.

Tricks

Trick: When only p, V and T change for a fixed mass, never compute n: use the ratio form $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$, and the units of p and V then cancel.

Trick: At constant volume, doubling the pressure doubles the Kelvin temperature, not the Celsius temperature.

10.2 Kinetic Theory and Molecular Speeds

Definitions & laws

- Assumptions of the kinetic theory of an ideal gas: (1) a gas consists of a very large number of identical molecules in continuous random motion; (2) the volume of the molecules is negligible compared with the volume of the container; (3) there are no intermolecular forces except during collisions; (4) collisions between molecules and with the walls are perfectly elastic; (5) the duration of a collision is negligible compared with the time between collisions; (6) Newton's laws of motion apply.
- **Pressure** of a gas is caused by the molecules colliding with the walls: the rate of change of momentum of the molecules per unit area of the wall.
- **Root-mean-square speed** c_{rms} is the square root of the mean of the squares of the speeds of the molecules.
- The average translational kinetic energy of a molecule is directly proportional to the thermodynamic temperature, and is independent of the type of gas.

- **Degrees of freedom** f is the number of independent ways in which a molecule can possess energy. Principle of equipartition of energy: each degree of freedom has an average energy of ½kT per molecule.
- A monatomic gas has f = 3 (translational), a diatomic gas has f = 5 (3 translational + 2 rotational) at ordinary temperatures, and a polyatomic (non-linear) gas has f = 6.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Pressure from kinetic theory	$p = \frac{1}{3}\rho\langle c^2\rangle = \frac{1}{3}(Nm/V)\langle c^2\rangle$	ρ = density (kg m^{-3}); N = number of molecules; m = mass of one molecule (kg); V = volume (m^3); $\langle c^2\rangle$ = mean square speed ($m^2\ s^{-2}$)
Root-mean-square speed	$c_{rms} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3p}{\rho}}$	M = molar mass (kg mol^{-1}); m = mass of one molecule (kg); T = temperature (K); k = Boltzmann constant (J K^{-1}); R = molar gas constant (J mol^{-1} K^{-1}); p = pressure (Pa); ρ = density (kg m^{-3})
Mean translational kinetic energy per molecule	$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	for the translational motion of one molecule (J)
Mean energy per molecule	$\frac{f}{2}kT$	f = number of degrees of freedom; k = Boltzmann constant (J K^{-1}); T = temperature (K)
Internal energy of an ideal gas	$U = \frac{f}{2}nRT = \frac{f}{2}NkT$	n = number of moles; N = number of molecules; f = number of degrees of freedom; U in J
Total translational kinetic energy	$\frac{3}{2}nRT$	n moles of gas; total translational kinetic energy in J; T = temperature (K)
Pressure and kinetic energy	$pV = \frac{1}{3}N\langle E_k\rangle$	N = number of molecules; $\langle E_k\rangle$ = mean translational kinetic energy per molecule (J)

How to answer (MPM style)

- For a derivation or statement of $p = \frac{1}{3}\rho\langle c^2\rangle$, list the assumptions first, then use the change of momentum per collision ($2mc_x$), the rate of collisions and $\langle c_x^2\rangle = \frac{\langle c^2\rangle}{3}$.
- For c_{rms}: write $c_{rms} = \sqrt{\frac{3RT}{M}}$, convert M to kg mol^{-1} and T to K, substitute and give the answer in m s^{-1} to 3 s.f.
- For a comparison of two gases at the same temperature: state that the average kinetic energy per molecule is the same, but $c_{rms} \propto \frac{1}{\sqrt{M}}$, so the lighter gas has the larger c_{rms}.
- Explain a rise in pressure when the temperature rises at constant volume: the molecules move faster, collide with the walls more frequently and with greater change of momentum per collision.

Section A traps

Trap: Stating that all molecules have the same speed; the molecules have a range of speeds, and only the rms (and mean) speed is quoted.

Trap: Quoting $\frac{3}{2}$kT as the internal energy per molecule of a diatomic gas: the translational part is $\frac{3}{2}$ kT, but the total energy per molecule is $\frac{5}{2}$kT.

Trap: Using the molar mass in g mol^{-1} in $c_{rms} = \sqrt{\frac{3RT}{M}}$, or doubling c_{rms} when T is doubled: c_{rms} increases by a factor of $\sqrt{2}$ only.

Tricks

Trick: $c_{rms} \propto \sqrt{\frac{T}{M}}$: doubling T raises c_{rms} by $\sqrt{2}$; at the same T the c_{rms} ratio of two gases is $\sqrt{\frac{M_2}{M_1}}$.

Trick: Same temperature means the same average translational kinetic energy per molecule, whatever the gas, so only the molecular mass decides the speed.